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CIRCULARITY IN THE DANISH MODERN CHAIR DESIGN

Analysis of three Danish modern classics with theories from Circular Economy (CE)

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ABSTRACT

The Danish furniture industry has since the mid twentieth century seen a steady rise in its popularity and demand, and this has been an international phenomenon (Hedemann & Nissen, 2013). This popularity is often attributed to the furniture designers of those times, like Børge Mogensen, Finn Juhl, Poul Kjærholm, Hans Wegner etc, whose designs from that era came to be known as the 'Danish modern furniture design' (Hansen, 2006; Teilmann-Lock, 2020).

Kaare Klint, often referred to as the father of Danish modern design, revolutionized early 20th-century furniture design education, where his pedagogy involved measuring and analysing human anatomy and traditional furniture to understand their function, enabling designers to create contemporary pieces by refining these 'types' to perfection (Hansen & Mussari, 2018; Teilmann-Lock, 2020, p. 5). One peculiar part of Klint's teachings was that his students were tasked with disassembling and measuring traditional chair types, such as Windsor or Chippendale chairs, to then design based on their analyses (ibid). As it so happened, majority of the designers from that era designed chairs, which would then go on to gain an 'iconic' status, between the period of 1930 to 1960 (Hansen & Mussari, 2018).

In the wake of new legislations aimed at promoting more sustainable and circular products to enter the EU market, amongst which 'furniture' is a product group that will see legislative mandates such as Digital Product Passports (DPP), new life-cycle analysis (LCA) methodologies like Product Environmental Footprint (PEF) etc which create the need for new knowledge (European Commission, 2020, 2022). In the context of Danish modern chairs, this knowledge needs to address whether the beloved designs from that era would fit into a new market where soon certain criteria will be imposed and whether these mandates would affect the designs in any way and more importantly, it would be interesting to see whether these iconic chairs can be called 'circular' or not.

Hence, this paper aims to explore how the design principles used in designing some of the iconic Danish modern chairs compare with the design principles from circular economy. This leads to the research question for this paper:

How do the design principles that defined the Danish modern chairs compare with design principles from Circular Economy (CE)?

To answer the research question, the paper will analyse three chairs of Danish modern design. The chairs have been selected based on their tectonics, which comprises of form, material composition/combination and construction method employed (De Gier & Buur, 2009).

This is a qualitative research paper (Justesen & Mik-Meyer, 2013) and will employ several methods from this research methodology. The analysis strategy for the Danish modern design principles will be based on five points, namely (a) the chair as an object, (b) material composition and combination, (c) the production process used, (d) design for repair, serviceability, take-back system etc and (e) value chain mapping, from the year of design to present day. The three chairs will be the subjects of a 'comparative case study' (Baxter & Jack, 2015) and the data will be collected from observation (Angrosino, 2009; Kawulich, 2005) of the selected chairs at the Royal Danish Academy's furniture collection. Content analysis (Krippendorff, 2004; Neuendorf, 2017) of documents such as sketches from the designers, production process images and videos from websites of companies producing these chairs etc will help will contribute to the overall analysis. Semi-structured interviews (Justesen & Mik-Meyer, 2013) with one or two practicing Danish furniture designers will further give insights into both the present-day scenario and the historical

perspective on Danish modern that these designers have grown up with. Diagrams as an analytical tool (Miles et al., 2014) will be crucial in rounding up the analysis and will inform the discussion.

Design principles from CE, namely (1) Design for disassembly (DfD), (2) Design for longevity, (3) Design for Reuse, (4) Design for Repair, (5) Design for Recycling, (6) Design for efficiency, (7) Design for biological cycles, (8) Design for Upgradability, (9) Design for sharing and (10) Design for dematerialisation will be used to analysed the data (Andrews, 2015; Earley, 2017; Ellen MacArthur Foundation, 2013b, 2013a; Haase et al., 2023; Kalmykova et al., 2018; Korhonen et al., 2018; Rizos et al., 2017).

The chairs in analysis are:



Figure 1



Figure 2



Figure 3

Note: The author will use photographs taken by them instead of these images, in the case that the abstract is accepted.

J46(fig. 1) spoke-back chair, designed by Poul M. Volther, an all-wooden chair from 1956 (FDB møbler, 2019). PK 22(fig. 2), lounge chair designed by Poul Kjærholm in 1956. It is made in brushed steel with wicker upholstery (Fritz Hansen, 2024). CH24 (fig. 3), also popularly known as the 'Wishbone chair' or the 'Y-chair' designed by Hans J. Wegner in 1949. It is a wooden chair with braided paper-yarn seat (Carl Hansen & Søn, 2024).

Preliminary analysis of the three chairs for this paper suggest that some of the design principles and practices from the Danish modern era have close resemblance to the design principles of CE such as design for longevity and design for repair, but with new criteria such as PEF LCA and DPP, some of the currently well-established practices will certainly be challenged. For instance, one of these practices refer to the craftsmanship that goes into making the organic wooden or metallic forms and how this affects the amount of material used, which in turn contributes to the LCA calculation of the chair, in terms of energy used to run the machines to shape the material. In light of the new legislative changes and CE principles, we might have to redefine or reframe our understanding of chair designs we have grown up studying, such as the Danish modern chairs.

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